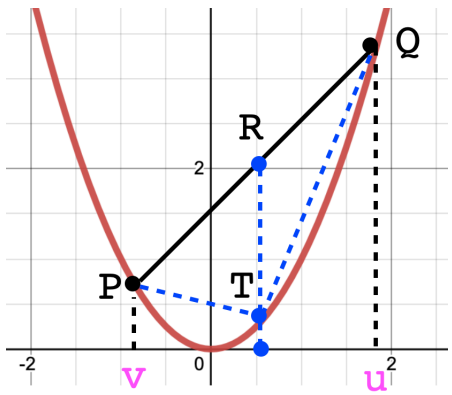
Archimedes' Quadrature

Quadrature refers to area, in particular to a method that can find the area of a plane figure. Consider the parabola $y = x^2$ with its vertex at the origin. For simplicity we will not worry about the shape factor a — showing that everything works with a included is left as an exercise.



Pick any two points on the parabola P and Q, then find the vertical line RT that lies exactly halfway in between.

Let the x-coordinates of Q and P be u and v, respectively. Then the y-coordinates of Q and P are u^2 and v^2.

Notice that RT produces two smaller triangles of equal area, since they have the common base RT and equal height. That height is half the horizontal distance between P and Q

$$h = \frac{u - v}{2}$$

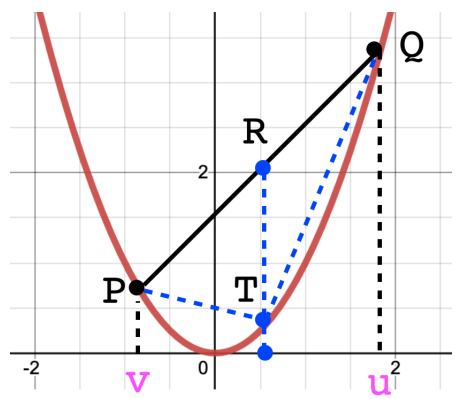
It is a moderate challenge to find the length of RT. The x-coordinates of the points R, T and the intersection of RT with the x-axis are all the same: the average of u and v or $(u+v)/2$.

Since T is also on the curve its y-coordinate is the square of the x-coordinate $[(u+v)/2]^2 = (u+v)^2/4$

The y-coordinate of R is the average of those for P and Q or $(u^2+v^2)/2$.

Finally, the length of RT is difference of those two y-coordinates:

$$b = \frac{u^2+v^2}{2} - \frac{(u+v)^2}{4}$$



Before proceeding to the area calculation, we simplify this expression.

$$4b = 2u^2 + 2v^2 - u^2 - 2uv - v^2$$

$$= u^2 - 2uv + v^2 = (u-v)^2$$

The area of each small triangle is $hb/2$ so the area of $\triangle PQT$ is

$$A = hb = \frac{u-v}{2} \cdot \frac{(u-v)^2}{4}$$

$$= \frac{1}{8}(u-v)^3$$

That's a pleasingly simple result, using only the x-values of the extreme points u and v.

We ask, what would happen if we repeat the process? Subdivide each of PT and TQ. Let w be the x-coordinate of the original bisector. Then the new distance is $u - w = (u - v)/2$.

$$A' \stackrel{?}{=} \frac{1}{8}(\frac{u-v}{2})^3$$

This is not quite right yet. The reason is that we have here the area for the two new triangles between, say, T and Q. But there are also two new triangles between P and T. So

$$A' = 2 \cdot \frac{1}{8}(\frac{u-v}{2})^3$$

$$A' = \frac{1}{4} \cdot \frac{1}{8}(u-v)^3$$

$$A' = \frac{1}{4}A$$

Repeating yet again would yield

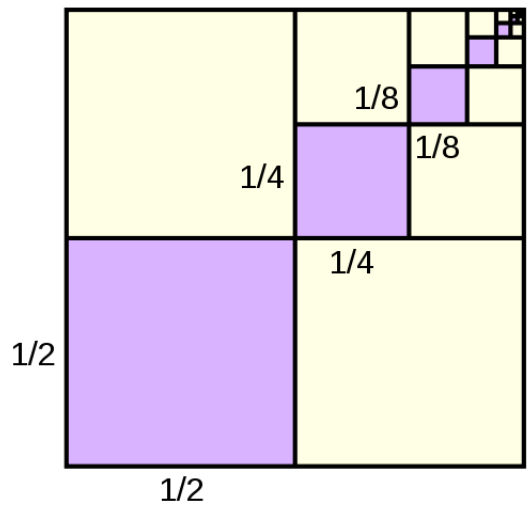
$$A'' = \frac{1}{4}A' = \frac{1}{16}A$$

Thanks to a method developed by Eudoxus, we carry out this until the area just above the curve is *exhausted*:

$$\Delta A = A' + A'' + \cdots =$$
$$= \ [\ \frac{1}{4} + \frac{1}{16} + \ldots\ ]\ A$$

This is a geometric series with $r = 1/4$ and first term $a = 1/4$ and its sum is known to be

$$\frac{a}{1-r} = \frac{1/4}{3/4} = \frac{1}{3}$$



so

$$\frac{A + \Delta A}{A} = \frac{4}{3}$$

The total area between the curve and the secant PQ is equal to 4/3 times the area of $\triangle PTQ$.

checking our work

We check this with a combination of simple calculus and not-so-simple algebra. The area below the curve is the integral:

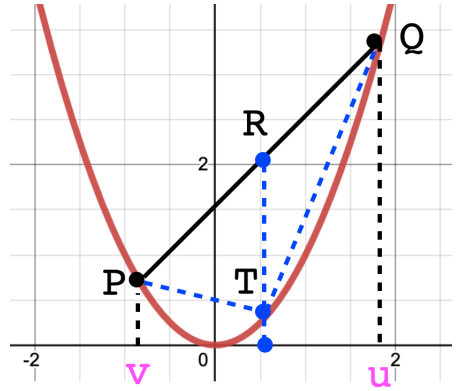
$$\int_v^u x^2 \, dx = \frac{1}{3}(u^3 - v^3)$$

We find a factor of $u - v$ in the above equation:

$$A_\int = \frac{1}{3}(u - v)(u^2 + uv + v^2)$$

Earlier, we determined that the $\triangle PTQ$ has area

$$\frac{1}{8}(u - v)^3$$



The trapezoid or box formed by PQ, the verticals, and the x-axis has for its average height the y-value of the point R:

$$h = \frac{u^2 + v^2}{2}$$

and width $u - v$ so its area is

$$A_{\text{box}} = (u - v) \cdot \frac{u^2 + v^2}{2}$$

The areas are

$$A_{\text{box}} = A_{\int} + \frac{4}{3} A_{\triangle}$$

We factor $u - v$ from each before re-writing the last equation as

$$\frac{u^2 + v^2}{2} = \frac{1}{3}(u^2 + uv + v^2) + \frac{4}{3} \cdot \frac{1}{8} \cdot (u - v)^2$$

Working with the right-hand side, we clearly we have a factor of 1/3, and that has me a little worried as there certainly is not one on the left.

$$= \frac{1}{3}(u^2 + uv + v^2 + \frac{(u - v)^2}{2})$$

We have from that second term $(u^2 - 2uv + v^2)/2$ which gives not only a cancelation but also factors of $3/2$:

$$= \frac{1}{3} \cdot \frac{3}{2}(u^2 + v^2) = \frac{u^2 + v^2}{2}$$

And that's a relief. ✓

halfway point

The slope of the tangent to the curve at point T is twice the x-coordinate of T, times a if present (see below). That's just $2 \cdot (u+v)/2 = u + v$.

The slope of the secant PQ from analytic geometry is the same:

$$\frac{\Delta y}{\Delta x} = \frac{u^2 - v^2}{u - v} = u + v$$

meeting tangents

It is a fundamental, simple result that the tangent line to the the curve $y = ax^2$ at a point (x, y) has slope $2ax$. We will derive this in a couple ways at the end. Here we have $a = 1$ so the slope is just $2x$.

Thus, the tangent through Q has slope $2u$ and we find the equation

$$y = 2ux + y_0$$

$$u^2 = 2u^2 + y_0$$

$$y_0 = -u^2$$

so

$$y = 2ux - u^2$$

similarly, the tangent through P is

$$y = 2vx - v^2$$

Let us consider the point M where the two tangents meet.

$$2ux - u^2 = 2vx - v^2$$

$$x = \frac{u^2 - v^2}{2(u - v)} = \frac{u + v}{2}$$

This is the x-value halfway between P and Q. The y-value is

$$M_y = 2u \cdot (u + v)/2 - u^2$$

$$= uv$$

The point T is on the curve so

$$T_y = (\frac{u + v}{2})^2 = \frac{(u + v)^2}{4}$$

The point R is halfway along the secant so

$$R_y = \frac{u^2 + v^2}{2}$$

Alternatively, the secant PQ has slope $u + v$ (see above)

$$y = (u + v)x + y_0$$

and it goes through Q so

$$u^2 = (u + v)u + y_0$$

$$y_0 = -uv$$

The equation of the secant is

$$y = (u + v)x - uv$$

So the value of y at $x = (u + v)/2$ is

$$R_y = \frac{(u + v)^2}{2} - uv = \frac{u^2 + v^2}{2}$$

So we have three points with the same x and corresponding y-values:

$$M_y = uv, \qquad R_y = \frac{u^2+v^2}{2}, \qquad T_y = \frac{(u+v)^2}{4}$$

$$\begin{aligned} R_y - T_y &= \frac{u^2+v^2}{2} - \frac{(u+v)^2}{4} \\ &= \frac{u^2}{4} - \frac{uv}{2} + \frac{v^2}{4} = \frac{(u-v)^2}{4} \end{aligned}$$

Similarly

$$\begin{aligned} T_y - M_y &= \frac{(u+v)^2}{4} - uv \\ &= \frac{(u+v)^2 - 4uv}{4} = \frac{(u-v)^2}{4} \end{aligned}$$

We have shown that

$$\begin{aligned} R_y - T_y &= T_y - M_y \\ 2T_y &= R_y + M_y \end{aligned}$$

parallel lines

In *Infinite Powers*, Strogatz describes this problem from the reverse direction. He says that Archimedes starts with the secant and then moves that line parallel to find where it just touches the curve.

The equation of the secant is

$$y = (u+v)x - uv$$

So if a point is on a line with that slope and is also on the curve $y = x^2$:

$$x^2 = (u+v)x - uv$$

$$x^2 - (u+v)x + uv = 0$$

This is a quadratic in x which has a single solution when $x = -b/2a$ (standard notation for the coefficients of the quadratic)

$$x = \frac{u+v}{2}$$

x lies halfway between u and v, as we said.

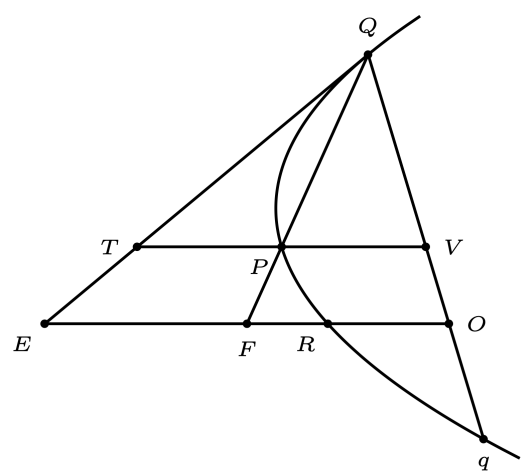
factor of a

A factor of a in the equation $y = ax^2$ stretches all y-values and all the areas in the y-direction. The simplest way to deal with this is to do a variable substitution $y/a = u$ and then follow what we did here.

In interpreting the areas, every area will have a factor of a to add back, from the stretching, and those will cancel. We leave that as an exercise.

second proof

Archimedes gave a second, quite novel proof of his theorem about the area of the parabola.



I have only been able to verify part of it.

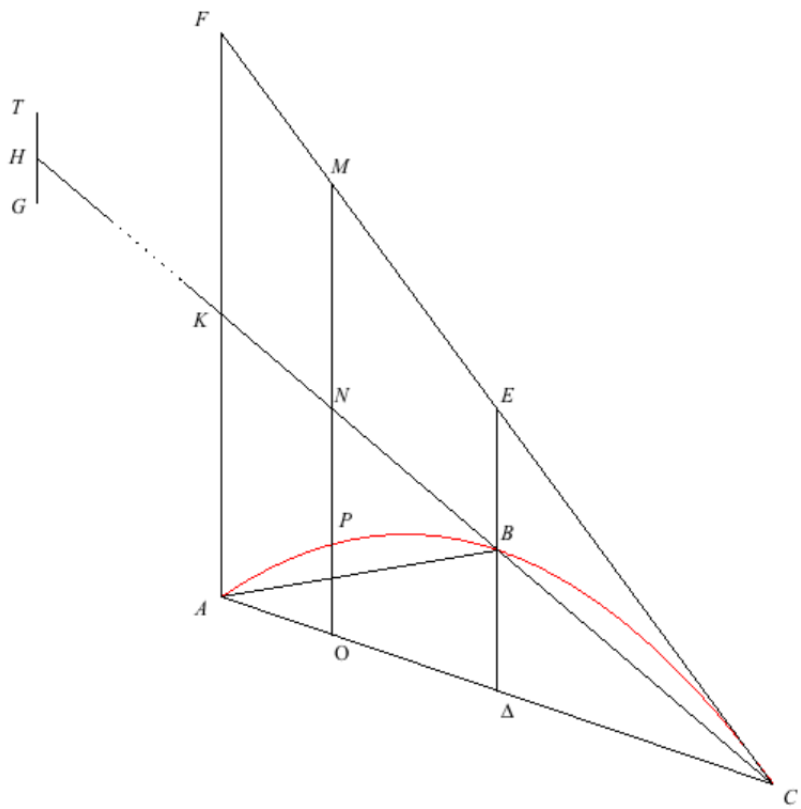
According to the notes I have

`http://math.fau.edu/yiu/Oldwebsites/MSTHM2014/2014MSTHMNotesH.pdf`

we need the following lemma: draw the tangent line to the curve at Q, and the midpoint of the line Qq. VP is drawn parallel to the axis of the parabola. Find where the extension of VP intersects the tangent line at T. Then TP is equal to PV.

In the notation of our original figure, we're extending RT to meet the tangent drawn through Q (not shown). The tangents (say, from Q) cuts the vertical through T into two equal parts, one is RT and the other RM. This is exactly what we proved above.

Archimedes says this is a "well known property", but I don't know how to solve it with pure geometry.



This figure has a Δ for what should be D. We have shown that $BE = BD$, and it is given that $AD = DC$ and AF, OM and DE are all vertical, so we have a bunch of similar triangles.

It follows that

$$A_{\triangle ACK} = 4 \cdot A_{\triangle DCB} = 2 \cdot A_{\triangle ABC}$$

$$A_{\triangle ACF} = 2 \cdot A_{\triangle ACK} = 4 \cdot A_{\triangle ABC}$$

second part of the proof

Drop the vertical MO. Then $AC : AO = MO : PO$. The second claim is that this is true *for any vertical*.

We rearrange the proportion:

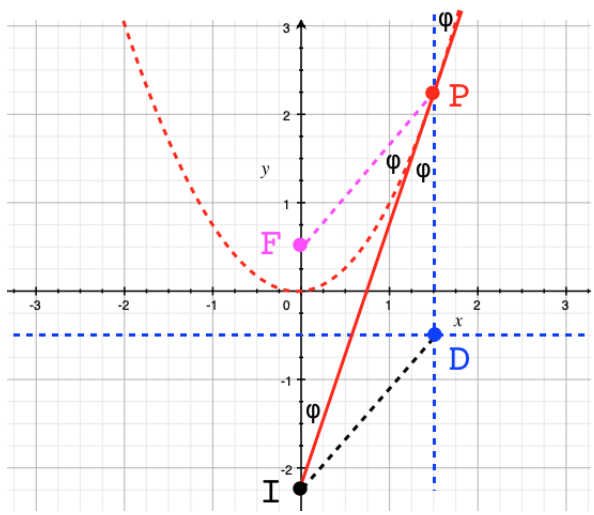
$$\frac{MO}{AC} = \frac{AO}{PO}$$

I do not know how this claim can be justified geometrically. I tried setting it up in coordinates but it turned into a mess.

slope of the tangent

At this point, you might ask why it is that if we pick two points and find the third point whose x is midway, that the slope of the secant through the two points is equal to the slope of the curve at x.

A French mathematician named Roberval had a beautiful answer for that.



The geometric definition of the parabola is to draw a line called the directrix and also pick a point F, the focus. Then every point on the parabola P has the same distance to F as vertically down to D on the directrix.

Since the direction of "motion" must keep these distances the same, it

follows that the tangent bisects $\angle FPD$ and then $FPDI$ is a parallelogram.

As a result the slope of the tangent line PI is $2ax^2/x = 2ax$.

A second approach comes from calculus. Literally the first thing we do there is to find the slope of the secant for two nearby points (separated by Δx), that satisfy a quadratic equation $y = ax^2$.

$$\begin{aligned} f'(x) &= \frac{a(x+\Delta x)^2 - ax^2}{\Delta x} \\ &= \frac{2ax\Delta x + (\Delta x)^2}{\Delta x} \\ &= 2ax + \Delta x \end{aligned}$$

The magic is that we can let Δx "tend" to zero (despite having it in the denominator the moment before), and then this becomes the slope of the tangent line, which is $2ax$.

A third approach is to write the equations for a line that intersects a parabola.

$$\begin{aligned} y &= kx + y_0 \\ y &= ax^2 \end{aligned}$$

So

$$ax^2 - kx - y_0 = 0$$

The roots are

$$x = \frac{k \ \pm \sqrt{k^2 - 4ay_0}}{2a}$$

There may be no solutions, one solution, or two. The case of one solution is the tangent line, and that happens when the discriminant is zero.

$$\begin{aligned} x &= \frac{k}{2a} \\ k &= 2ax \end{aligned}$$

focus

One last thing is to find the focus and work with it briefly. The focus is the point $F = (0, f)$ with the property that for every (x, y) on the parabola, the distance to the directrix $y = -f$ is

$$PD = ax^2 + f$$

and that is equal to the distance to the focus.

$$PF = \sqrt{(ax^2 - f)^2 - x^2}$$

Thus

$$(ax^2 + f)^2 = (ax^2 - f)^2 - x^2$$
$$2ax^2 f = -2ax^2 f - x^2$$
$$4af = 1$$

Here we just have $a = 1$ so $f = 1/4$. The focus F is the point $(0, 1/4)$.

If we pick any points $P = (v, v^2)$ and $Q = (u, u^2)$, the line between them has slope

$$\frac{u^2 - v^2}{u - v} = u + v$$

and equation

$$y = (u + v)x + y_0$$
$$u^2 = (u + v)u + y_0$$

so

$$y = (u + v)x = uv$$

If this line also goes through F, then $y_0 = uv = f = -1/4$ Multiply the slopes of the tangent lines

$$2u \cdot 2v = 4uv = -1$$

We conclude that if the line connecting P and Q goes through the focus F, then the tangents through those points are perpendicular.

Here's a geometric proof from Kline.

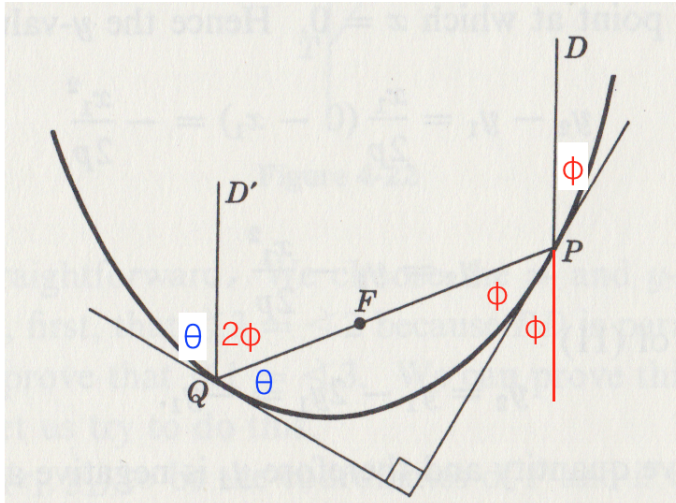


Figure 4-23

This relies on the fact that the line from the focus to any point on the parabola forms the same angle with the tangent as does the vertical. This is the the so-called "headlight" property of the parabola.

Alternatively, you can accept the algebraic proof and use this as a proof of the headlight property.